Final Year Project Report

Event-Guided Diffusion Transformer for Video Frame Interpolation in Dynamic Scenario

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Abstract

Video Frame Interpolation (VFI) often struggles in dynamic environments where rapid object movement causes severe motion blur and temporal aliasing. Standard shutter-based cameras lack the necessary temporal density between exposures to reconstruct non-linear trajectories accurately. To address these limitations, this thesis proposes an Event-Guided Diffusion Transformer framework. By utilizing the microsecond-level motion evidence from event cameras, the model gains precise temporal anchors to track motion that is otherwise invisible to traditional frame-based sensors. Our framework integrates a Multi-Head Cross-Attention (MHCA) mechanism within a Diffusion Transformer (DiT) backbone to align asynchronous event features with spatial latent patches. Furthermore, we introduce a motion-aware training strategy that employs a weighted loss function to prioritize the reconstruction of dynamic regions over static backgrounds. Experimental results on the Middlebury and SNU-FILM benchmarks demonstrate that our approach significantly outperforms several state-of-the-art models. Specifically, our method reduces perceptual error by up to 21.4% and improves temporal consistency by up to 27.3% compared to diffusion-based baselines like LDMVFI. Qualitative evaluations confirm that the proposed model effectively suppresses "ghosting" artifacts and maintains sharp structural boundaries in extreme motion scenarios.

Key words: Video Frame Interpolation, Diffusion Transformer, Event Camera, Generative Artificial Intelligence, Computer Vision

Table of Contents

1. Introduction

1.1 Research Background and Significance

Video Frame Interpolation (VFI), functions as a foundational pillar within the broader scientific pursuit of temporal super resolution. It is specifically engineered to synthesize coherent and visually consistent intermediate frames that reside between sparse anchor images, thereby facilitating the reconstruction of a continuous motion stream. This technique is vital for circumventing the inherent physical and mechanical sampling constraints that characterize traditional imaging hardware. Such a capability is fundamentally instrumental in bridging the significant technological chasm that exists between the discrete temporal captures of electronic sensors and the seamless, fluid dynamics observed in the actual physical world. Historically, this technology was viewed as a highly specialized post production utility designed for cinematic editing or niche artistic visual effects. However, it has since undergone a significant paradigm shift, maturing into a core architectural component that underpins a diverse and expanding array of high-speed computational vision frameworks [1].

In the contemporary and rapidly advancing landscape of high-performance consumer electronics, the widespread proliferation of sophisticated high refresh rate displays has introduced a formidable set of technological challenges that require immediate attention from the computational vision community. Modern monitors and mobile device screens now frequently feature hardware capabilities exceeding 120 hertz or even 144 hertz, providing the potential for exceptionally fluid visual experiences that were previously unattainable. Nevertheless, a significant technological gap persists because the vast majority of available digital content remains captured and transmitted at standard frame rates that are significantly lower, typically hovering around 24 or 30 frames per second. Video Frame Interpolation (VFI) serves as the primary and most effective solution to this systemic disparity by facilitating the up conversion of standard video footage into high frequency sequences through the mathematically complex and intelligent synthesis of intermediate temporal data. By operating as a bridge between the limitations of content production and the capabilities of modern hardware, this technology ensures that the visual output is commensurate with the display specifications [2].

This sophisticated process effectively mitigates the presence of perceivable motion judder and stuttering, which in turn enhances the subjective fluidity and the overall depth of visual immersion for the observer. The human visual system is highly sensitive to the discrete jumps inherent in low frame rate content when presented on high-speed displays, a phenomenon often referred to as the sample and hold effect. By generating visually coherent intermediate frames, VFI reduces the retinal slip and provides a much smoother transition between states, thereby creating a more naturalistic viewing experience. Furthermore, VFI provides a powerful and indispensable toolset for the digital media and sports broadcasting industries by allowing for the creation of professional grade and ultra slow-motion effects using standard recording hardware. In sports broadcasting, for instance, the ability to observe the minute details of high velocity movements is essential for both officiating and audience engagement. This democratization of high-speed cinematography is particularly vital in professional environments where the deployment of native high-speed cameras is either cost prohibitive or technically limited by environmental factors. High speed sensors often require significant illumination to maintain short exposure times, which is not always feasible in outdoor or low light settings. Additionally, the massive data rates produced by native high frequency cameras often exceed the storage capacity and bandwidth of mobile production units, making the algorithmic synthesis of frames a more viable and scalable alternative [2].

Parallel to its substantial aesthetic and commercial utility, Video Frame Interpolation has evolved into an indispensable requirement for the safe operation of safety critical autonomous systems, which include both self-driving vehicles and industrial robotic platforms. Standard sensors that utilize a shutter-based mechanism are fundamentally limited by their integration time, which is a physical constraint that inevitably results in significant blind periods between each discrete exposure. During these critical micro intervals, the sensor is essentially inactive, meaning that rapid environmental changes or high velocity object movements can occur completely undetected by the primary perception stack. In a high-speed driving scenario, a vehicle may travel several meters during the time it takes to capture two consecutive frames, creating a dangerous information gap. These information voids frequently precipitate severe temporal aliasing, which can manifest as the incorrect estimation of object velocity or direction, leading to catastrophic tracking failures in high stakes environments where accuracy is paramount for collision avoidance [3].

By restoring the necessary temporal density through the rigorous process of frame interpolation, these autonomous systems ensure that machines can maintain a persistent, reliable, and continuous understanding of their dynamic surroundings. The goal is to move beyond the limitations of discrete snapshots and achieve a state of continuous environmental awareness. As digital perception requirements continue to trend toward higher precision and greater reliability, the ability to reconstruct inter frame motion that is physically grounded has evolved from a secondary optimization into a rigorous technical necessity. This evolution is driven by the need for autonomous agents to make split second decisions based on the most current and accurate motion evidence available. Reconstructing the true trajectory of an object rather than relying on linear approximations is essential for achieving accurate environmental awareness and ensuring operational safety in high-speed scenarios. Consequently, the development of robust VFI frameworks that can handle the complexities of nonlinear motion and varying lighting conditions remains a core objective for the next generation of digital perception systems [3].

1.2 Current Research Status

1.2.1 Evolution of video frame interpolation

The historical trajectory of Video Frame Interpolation (VFI) signifies a fundamental and transformative paradigm shift, transitioning from the era of classical signal processing techniques characterized by handcrafted rules to the modern landscape of sophisticated architectures driven by data. This evolution represents a departure from the manual engineering of features toward the automatic discovery of complex temporal patterns through large scale optimization. In the period that preceded the dominance of deep learning, the prevailing methodologies predominantly relied on shifting mechanisms based on phase or heuristics for the matching of blocks to establish temporal correspondences between discrete images. These techniques were rooted in the assumption that motion could be modeled through deterministic mathematical transformations of pixel intensities, a framework that served as the primary tool for temporal resolution enhancement for several decades [1].

By conceptualizing the task as a localized problem of pixel displacement, these early approaches attempted to extrapolate linear motion trajectories between consecutive frames by utilizing the mathematical framework provided by the optical flow constraint equation. This equation assumes that the brightness of a moving point remains constant over a very short temporal interval, allowing for the derivation of motion vectors that describe the movement of pixels. While these solutions offered significant computational tractability, a quality that made them highly suitable for implementation in hardware systems operating in real time, they were notoriously susceptible to environmental perturbations and fluctuations in scene luminance. The fragility of the brightness constancy assumption meant that even minor changes in lighting or the presence of sensor noise could lead to catastrophic failures in motion estimation. Furthermore, because these methods lacked the benefit of robust semantic priors, they often suffered from severe artifacts, such as structural tearing or ghosting, when they were confronted with complex textures or surfaces that do not follow Lambertian properties. The inability to distinguish between true object motion and specular reflections or shadows limited their applicability in diverse and unpredictable real-world environments [1].

The advent of deep learning catalyzed the development of significantly more resilient and integrated frameworks designed from end to end, which can be broadly categorized according to their specific strategies for the modeling of motion. Among these, paradigms based on flow, which are exemplified by the seminal and foundational work of Huang et al. [1], continue to serve as a cornerstone of contemporary research in this field. These architectures are designed to explicitly estimate the underlying fields of optical flow, which are then used to warp pixels from the source frames into the target intermediate timestamp. This explicit modeling of movement allows the network to prioritize geometric consistency and structural alignment, ensuring that the synthesized content remains faithful to the spatial layout of the original scene. By treating the problem as a warping task, these models can effectively preserve the high frequency details of the input frames while maintaining a coherent global structure through the use of sophisticated encoder decoder pipelines.

To further address the significant challenges posed by occluded regions where the mapping between frames becomes inherently ambiguous or bidirectional, researchers have integrated reasoning that is aware of depth to prioritize the warping of foreground elements over background content [4]. In a typical video sequence, objects moving at different distances from the camera create complex occlusion patterns that cannot be resolved through simple flow estimation alone. By incorporating depth as a guiding prior, these frameworks can determine the relative visibility of pixels, ensuring that foreground objects correctly occlude the background during the synthesis of the intermediate frame. This hierarchical approach to motion modeling provides a much more physically plausible reconstruction of the scene dynamics, effectively reducing the artifacts that typically occur at the boundaries of moving objects. This transition toward architectures that are aware of semantics and geometry highlights the ongoing effort to overcome the fundamental bottlenecks of traditional videography based on frames.

Conversely, kernel-based methods, such as those pioneered by Lee et al.[5], forgo the rigid constraints of explicit pixel warping in favor of localized synthesis. Instead of moving pixels, these models focus on predicting spatially-adaptive, multi-scale kernels that are convolved with input frames to generate new pixel values. This approach provides superior flexibility in blending information from neighboring regions, particularly in scenarios where flow estimation is inherently unreliable due to sudden motion or blur. The effectiveness of this strategy was significantly enhanced by the introduction of separable convolution layers, which allow for larger receptive fields without the exponential increase in computational cost [6].

In more recent years, the research focus has migrated toward Transformer-based architectures, such as the framework proposed by Lu et al. [7], which treat interpolation as a global sequence-modeling task. By leveraging multi-head self-attention mechanisms, these models capture long-range spatio-temporal dependencies that traditional convolutional layers, restricted by a fixed receptive field, often fail to perceive. Despite their state-of-the-art performance on standardized benchmarks, these models remain essentially bottlenecked by the quality and sampling frequency of the input RGB signal. Their fundamental reliance on visually clear frames renders them ineffective in high-speed scenarios where rapid object movement triggers severe motion blur and temporal aliasing. In such dynamic environments, the critical motion evidence is often "smeared" or permanently lost between exposures, leading to the structural distortions and "ghosting" artifacts that plague purely frame-based systems.

1.2.2 Limitations in dynamic scenarios

The computational architectures previously detailed demonstrate a significant performance degradation when they are deployed in high dynamic environments. This notable decline in efficacy is primarily attributed to the inherent sampling limitations and fixed integration periods of standard frame-based sensors when they are tasked with capturing rapid scene changes. In these volatile and unpredictable settings, the confluence of rapid object motion and high velocity camera ego motion generates profound motion blur and debilitating temporal aliasing effects that fundamentally obscure the underlying scene geometry. When the spatial displacement of pixels between consecutive RGB frames transcends the effective receptive field of the neural network, the conventional assumptions of linear motion and constant brightness are fundamentally violated. These core assumptions, which serve as the foundation for the vast majority of traditional interpolation algorithms, rely on the premise that movement between frames is both predictable and uniform. However, this simplistic idea collapses entirely when faced with the stochastic complexities of real-world acceleration or non-linear paths where objects do not follow a straight or consistent trajectory.

As rigorously analyzed by Tulyakov et al. [8], these adverse conditions precipitate ghosting artifacts and severe structural distortions, which compromise both the perceptual quality and the mathematical integrity of the synthesized frames. The presence of motion blur effectively removes the high frequency textural details required for precise feature matching and correspondence estimation, leading to an under determined optimization problem where the neural network must essentially hallucinate the missing informational content. Furthermore, temporal aliasing introduces significant ambiguities in the motion field, as the discrete sampling rate of the camera fails to satisfy the necessary criteria for capturing high velocity dynamics without distortion. This results in the generation of visually inconsistent content where moving edges appear smeared or duplicated across the temporal axis, a failure that highlights the critical need for a more robust sensing paradigm. In environments where precision is a technical necessity, such as autonomous navigation or high-speed industrial monitoring, these structural failures are not merely aesthetic issues but represent a fundamental breakdown in the reliability of the digital perception system. By failing to capture the microsecond level motion evidence occurring during the inter-exposure intervals, purely frame based models remain trapped by the physical constraints of their hardware, necessitating the integration of auxiliary asynchronous data to restore the lost motion continuity.

The root cause of this systemic failure lies in the operational principle of standard shutter-based sensors. These traditional devices are effectively blind during the inter-exposure intervals, meaning they fail to capture the necessary motion evidence to accurately reconstruct non-linear or high acceleration trajectories. Because standard cameras act as discrete samplers, any information occurring between the moments the shutter is open is permanently lost to the system. This creates an under determined problem where the algorithm must essentially guess the state of the scene during these blind periods. Furthermore, the integration of light over a fixed exposure time in high-speed scenarios results in the spatial smearing of features, which removes the high frequency textural details required for precise feature matching and temporal alignment.

To mitigate these intrinsic hardware level limitations, this research integrates the high frequency temporal data provided by event cameras. Unlike conventional CMOS sensors that sample global intensity at fixed periodic intervals, event-based sensors operate asynchronously, recording per pixel log intensity changes with microsecond level latency. This bio inspired sensing paradigm yields a continuous, high temporal resolution record of scene dynamics, effectively filling in the motion that occurs during the traditional camera's blind spots. By leveraging these sparse but temporally dense signals, the proposed framework establishes a precise temporal anchor. This anchor serves as a continuous physical record of change rather than a set of discrete snapshots [9].

This temporal anchor provides the requisite physical evidence to track and reconstruct complex motion paths that remain entirely invisible to frame based systems, ensuring that the synthesized intermediate frames are both visually sharp and physically consistent with the true underlying motion. The logarithmic response of these sensors further enables the system to maintain a high level of sensitivity even in extreme lighting conditions, providing a robust stream of motion evidence where traditional sensors would fail due to overexposure or underexposure. By aligning this asynchronous data with the latent representations of the diffusion transformer, the model can successfully resolve motion ambiguities and restore the sharp structural boundaries of fast-moving objects, thereby overcoming the fundamental sampling bottlenecks of traditional videography [10].

1.2.3 Conflict between pixel-wise accuracy and visual realism

A persistent and profound challenge within the scientific field of Video Frame Interpolation is the historical and pervasive reliance on objective pixel level metrics, most notably the Peak Signal to Noise Ratio and the Structural Similarity Index. These metrics have traditionally served as the primary benchmarks for evaluating the performance of interpolation algorithms due to their mathematical simplicity and their ability to provide a reproducible framework for the quantitative comparison of diverse models. While these specific metrics provide a mathematically convenient and highly reproducible framework for objective evaluation, they are increasingly subject to rigorous criticism due to their demonstrably poor correlation with subjective human visual perception. The human visual system is highly sensitive to structural integrity, edge sharpness, and textural consistency, all of which are qualities that are often not fully captured by the simple calculation of average pixel differences or local intensity correlations.

As meticulously observed and documented by Danier *et al*. [11], optimization strategies that prioritize the minimization of pixel wise loss functions, such as the Mean Squared Error, often yield synthesized frames that appear perceptually soft or significantly blurred to the human eye. This lack of visual sharpness is especially prevalent in regions of the image where motion is particularly fast or unpredictable, as the model struggles to determine the exact spatial coordinates of moving objects. This undesirable phenomenon is understood to be a direct and unavoidable consequence of the perception distortion tradeoff, which serves as a foundational principle in signal processing and generative modeling. When a model is faced with the inherent and substantial uncertainty associated with high velocity motion, it encounters a multimodal probability distribution regarding the potential location of a pixel. To minimize the expected error and avoid the high penalties associated with large spatial mistakes, the algorithm typically predicts a statistical average of all potential locations. By choosing the mean of the probability distribution, the model effectively minimizes the overall variance but produces a result that lacks the high frequency components required for a sharp and realistic image. Consequently, while such models based on regression may achieve superior scores on traditional benchmarks by avoiding large mathematical errors, they systematically sacrifice the critical high frequency components, such as sharp boundaries and intricate textures, that the human visual system prioritizes when assessing the realism of a digital scene [12].

To bridge this significant gap between mathematical proximity and visual realism, recent advancements in the field of generative modeling have emerged as a powerful and promising alternative to deterministic regression. In particular, the frameworks based on motion aware latent diffusion proposed by Huang et al. [13] have demonstrated a remarkable and robust capacity for synthesizing complex or highly ambiguous motion patterns that frequently cause traditional methods to fail completely. Unlike conventional architectures that perform a deterministic and often incorrect pixel wise guessing game, these generative models learn to sample directly from the underlying data distribution. This stochastic approach enables the reconstruction of high fidelity details and realistic textures that closely mirror the quality of real-world video sequences, as the model can generate a single sharp outcome rather than a blurry average of multiple possibilities. This paradigm shift in the approach to temporal super resolution significantly informs our strategic decision to adopt the Diffusion Transformer architecture [14], which offers superior scalability and the benefit of global spatial reasoning when compared to the local and restricted receptive fields of traditional convolutional backbones. The self-attention mechanisms within the transformer allow for the modeling of long-range spatial dependencies, which is essential for maintaining global consistency across the interpolated frame. By synergistically coupling the generative power of latent diffusion with the microsecond level physical motion evidence provided by sensors based on events, our proposed framework avoids the safe over smoothing characteristic that has long plagued prior interpolation approaches. Our ultimate objective is the synthesis of intermediate frames that are both mathematically reliable and visually sharp, maintaining the structural integrity of the scene even in the most demanding and dynamic environments [15].

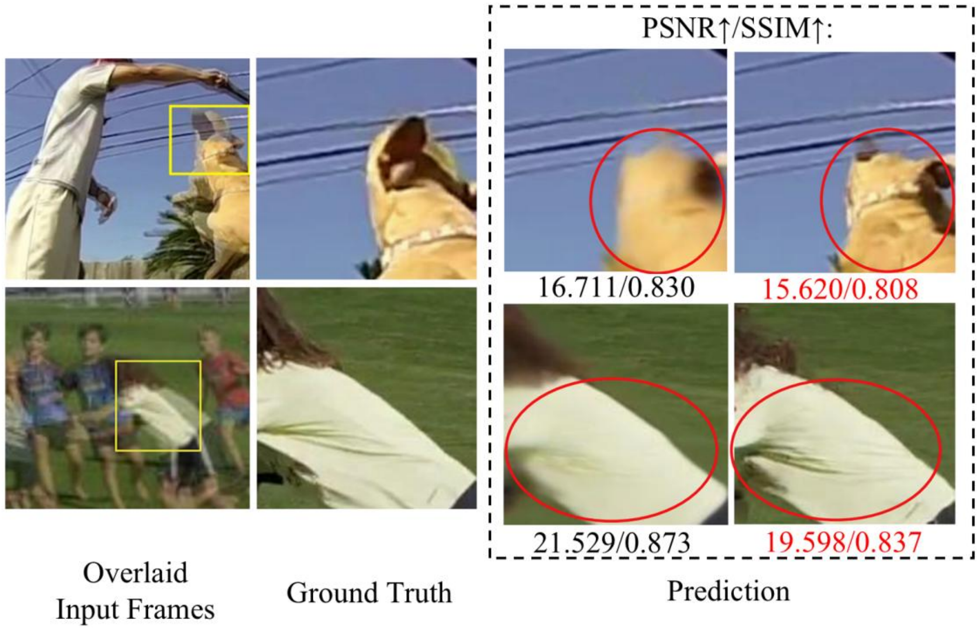

Fig 1. Visual illustration of the inconsistency between PSNR/SSIM and visual quality.

1.3 Main Contributions

The main contributions of this report are as follows:

(1)The proposed framework significantly enhances visual realism and temporal consistency, reducing perceptual error by up to 21.4% and improving consistency by up to 27.3% compared to the diffusion-based baseline.

(2)We introduce a novel Diffusion Transformer that leverages event signals to guide high-speed motion reconstruction, effectively addressing the motion ambiguities that often compromise RGB-only frameworks.

(3)We design a multi-objective motion-aware loss function that increases training efficiency and improves the model’ s ability to handle complex, high-speed movements in dynamic scenes.

2. Related Work

2.1 Event-based Video Frame Interpolation

To circumvent the inherent temporal bottlenecks of conventional shutter-based sensors, recent research has pivoted toward the integration of event cameras, which are bio-inspired vision sensors that detect per-pixel brightness changes asynchronously [16]. These sensors offer a significant advantage by capturing high-speed motion evidence during the traditionally "blind" intervals between consecutive RGB frames, providing a dense temporal bridge for reconstruction [17]. The technical challenge lies in effectively representing these sparse, asynchronous events for use in deep neural networks, leading to the development of sophisticated spatio-temporal representations such as voxel grids and event surfaces [18].

A pioneering contribution to this domain was the Time Lens framework introduced by Tulyakov et al. [8]. This architecture employs a dual-branch strategy, combining flow-based warping to preserve geometric structures with a synthesis-based refinement branch to handle complex occlusions and non-linear motion trajectories. This methodology was later expanded into Time Lens++ [19], which incorporates parametric non-linear flow and multi-scale feature fusion to achieve higher precision in predicting intermediate state transitions. Building on these foundations, subsequent researchers have explored multi-scale recursive architectures to better align event-driven motion priors with the high-resolution spatial details of RGB frames [20].

Despite the impressive performance gains of these hybrid models, they frequently struggle with artifacts in low-texture regions where the event signal may be too sparse to provide reliable motion cues. Because many of these frameworks rely heavily on traditional pixel-warping operations, they are prone to producing "ghosting" artifacts or structural tearing when the underlying event stream is noisy or incomplete. More recent attempts have sought to mitigate these issues by utilizing high-frequency guidance from event signals to steer the generation process [21]. Furthermore, the introduction of direct synthesis methods with event-based references, such as the work by Liu et al. [22], has demonstrated that leveraging events as a structural anchor can significantly improve visual realism in difficult scenarios. However, the conflict between maintaining pixel-wise accuracy and achieving high-fidelity textures in low-contrast environments remains a critical research gap, necessitating a more robust generative approach to maintain physical consistency [23].

2.2 Diffusion Models for Video Frame Interpolation

Recent breakthroughs in generative artificial intelligence have catalyzed a fundamental shift in Video Frame Interpolation from traditional regression-based frameworks to probabilistic generative modeling. Unlike conventional models that treat VFI as a deterministic mapping between frames, diffusion-based approaches model the underlying distribution of video data to synthesize more realistic content. This paradigm shift prioritizes perceptual fidelity over mere pixel level proximity, effectively mitigating the over-smoothing artifacts and blurry edges typical of architectures optimized solely for Mean Squared Error. The foundational work in video diffusion models by Ho et al. [24] established the feasibility of utilizing denoising probabilistic processes to capture the complex temporal dynamics inherent in video sequences, paving the way for specialized interpolation tasks.

Building upon these generative foundations, Danier et al. [11] introduced the LDMVFI framework, demonstrating that performing the diffusion process within a compressed latent space significantly enhances the generation of high-resolution frames while maintaining superior visual quality. This effectiveness was further validated by Jain et al. [25], who proved that diffusion models possess the inherent capacity to resolve highly non-linear motions that typically baffle traditional flow based or kernel-based frameworks. To address the issue of temporal artifacts in dynamic environments, Huang et al. [13] introduced motion aware priors to ensure that the synthesized content remains temporally coherent even when dealing with stochastic dynamic textures. Additionally, the development of masked conditional video diffusion (MCVD) by Voleti et al. [26] showcased the versatility of diffusion models in various temporal tasks, including both interpolation and extrapolation, by leveraging block conditioning techniques.

Nevertheless, a critical limitation persists in current diffusion based VFI research: a heavy reliance on standard RGB frames. In scenarios characterized by high velocity motion, the substantial displacement between consecutive frames results in severe information loss due to sensor level motion blur and temporal aliasing. These RGB only frameworks lack the dense temporal evidence required to accurately localize fast moving objects during the inter-frame "blind" periods. Consequently, while the generated results may satisfy perceptual realism, they often lack strict physical accuracy. The emergence of the Diffusion Transformer (DiT) as proposed by Peebles and Xie [14] offers a solution through its exceptional scalability and its ability to process diverse conditional inputs via global attention mechanisms. By integrating high speed, asynchronous event streams as a conditional guiding signal within a DiT based architecture, as explored in recent conditional control studies [27], there exists a unique opportunity to achieve a synthesis of visual sharpness and physical reliability. This integration leverages the microsecond level motion evidence of event cameras to steer the DiT's generative process [28], ensuring that the reconstructed motion remains grounded in physical reality even in the most demanding dynamic scenarios [29].

3. Methodology

3.1 Preliminary: Diffusion Transformer

The Diffusion Transformer, often referred to as DiT, signifies a fundamental architectural departure from the long-standing dominance of architectures based on the U-Net structure within the domain of latent diffusion models. While traditional U-Net models rely heavily on structural inductive biases such as spatial locality and hierarchical downsampling through convolutional layers, the DiT framework prioritizes a more scalable and flexible design based on the Transformer. As originally conceptualized by Peebles *et al.* [14], this architecture capitalizes on the massive successes of the Vision Transformer by treating the denoising process as a sequence modeling task rather than a purely spatial translation. By replacing the convolution centric layers with attention centric blocks, the model gains the ability to process visual information without the structural constraints of fixed receptive fields [30].

The operational workflow of the DiT begins with the representation of data within a compressed latent space. Typically, an input image is first encoded into a latent vector via a pre-trained Variational Autoencoder to reduce computational complexity. The DiT then partitions these latent features into a sequence of discrete patches, effectively transforming a two-dimensional spatial grid into a one-dimensional array of tokens. Each patch is linearly projected and combined with positional embeddings to preserve spatial relationships within the Transformer blocks. This process is crucial because it allows the model to treat image regions as independent entities that can interact globally through the self-attention mechanism, facilitating the capture of long-range dependencies that often elude traditional convolutional networks.

At its core, the DiT architecture consists of a sequence of standardized Transformer blocks, each incorporating multi head self-attention and pointwise feedforward networks. During the denoising phase, these blocks are tasked with approximating the noise distribution or the score function of the data at various diffusion timesteps. The self-attention layers enable every patch to attend to every other patch, providing a global receptive field from the very first layer of the network. Furthermore, the feedforward networks apply non-linear transformations to each patch independently, enhancing the capacity of the model to learn complex visual features. This modular design allows for highly efficient scaling, where increasing the number of blocks or the hidden dimension allows the representational power to grow predictably in accordance with scaling laws observed in large models [31].

One of the most advantageous characteristics of the DiT is its inherent flexibility in handling diverse conditional inputs. Beyond simple time step embeddings, the architecture can seamlessly integrate conditioning information such as class labels, text prompts, or high frequency event signals through specialized mechanisms like cross-attention layers or adaptive layer normalization. The cross-attention mechanism allows the latent patches to query external conditional features, ensuring that the generated content is strictly aligned with the provided guidance. This adaptability makes the DiT a highly potent framework for complex visual synthesis tasks, particularly in scenarios that require the fusion of multimodal data to maintain physical consistency and high-fidelity detail.

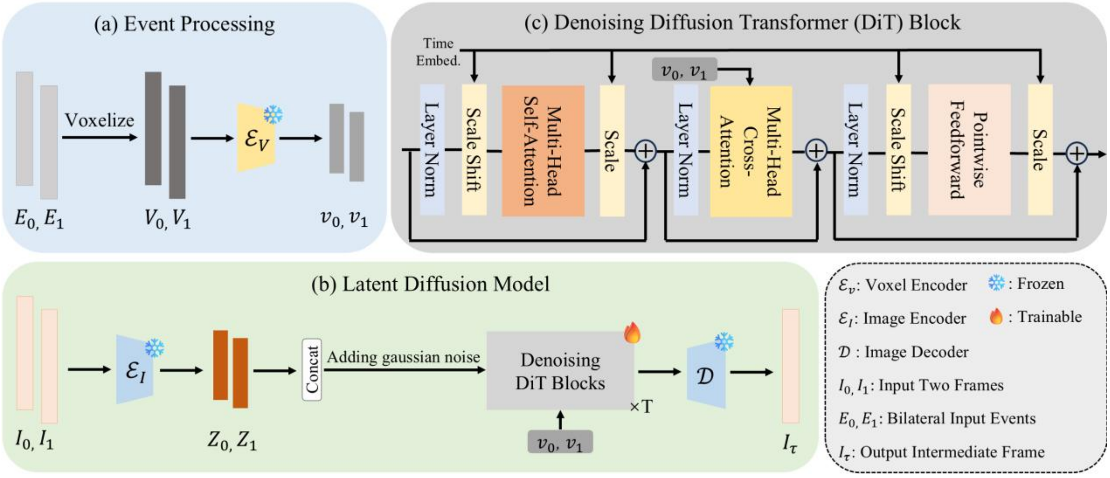

Fig 2. Overview of the proposed framework.

3.2 Event Representation

An event stream is fundamentally defined as a continuous and asynchronous sequence of neuromorphic signals that are triggered at the individual pixel level in response to dynamic changes in scene brightness. Unlike the global sampling approach of traditional cameras, this sensing paradigm operates independently for each pixel, where each discrete event e_i is meticulously characterized by a multidimensional tuple. This tuple includes the precise spatial coordinates (x_i, y_i) located on the sensor manifold, a binary polarity value $p_i \in \{-1, +1\}$ that indicates the specific direction of the logarithmic intensity shift, and a microsecond level timestamp t_i that captures the exact moment of the trigger. Because these events are only generated when a significant change in radiance occurs, the resulting raw data is characterized by extreme sparsity and a highly irregular sampling rate in the temporal domain. Such a non uniform and sparse structure presents a fundamental challenge for modern digital perception systems, as it is inherently incompatible with the rigid, grid-based tensor formats that define standard convolutional neural networks and transformer architectures. To effectively bridge this significant structural gap and allow for the application of advanced deep learning techniques, a frequently utilized approach [8], as visually represented in Figure 3(a), involves the systematic discretization of the temporal dimension into \$B\$ consecutive bins. This procedure facilitates the integration of the sparse event stream into a structured and dense three-dimensional spatiol temporal Voxel Grid $V \in \mathbb{R}^{\mathbb{B}\times\mathbb{H}\times\mathbb{W}}$. By accumulating the asynchronous polarity signals within these fixed temporal intervals, the transformation effectively converts the raw point cloud like data into a regular volumetric representation. This structured lattice is critical because it preserves the essential temporal distribution and the high frequency motion characteristics of the original signal while simultaneously enabling the use of high throughput architectural backbones for robust feature extraction and complex visual reasoning. The accumulation of events into a specific temporal bin k is performed using a linear interpolation kernel, which preserves the temporal distribution of the signals. This process is formulated as follows:

$$V(k) = \sum_{i=1}^{M} p_i\, max\left(0,1 - \left|k - \frac{t_i - t_s}{t_e - t_s}(B-1)\right|\right), \tag{1}$$

where t_s and t_e represent the start and end times of the integrated window, respectively, and M denotes the total number of events within that interval. The bin index k ranges from 0 to $B-1$. In our framework, we apply this method to the bilateral event streams E_0 and E_1 surrounding the target intermediate frame timestamp, generating two distinct voxel grids V_0 and V_1. As depicted in Fig. 3(a), to extract high-level motion features from these grids, we adopt the Voxel Encoder $\mathcal{E}_V$ from Liu *et al.* [22]. By keeping the encoder frozen, the proposed framework leverages robust, pre-learned motion priors to generate the condition features v_0 and v_1 . The transformation functions as a sophisticated computational bridge that systematically converts the raw and high frequency temporal dynamics of the asynchronous event stream into a series of structured and dense feature tensors. By mapping these sparse neuromorphic spikes into a regularized latent manifold, the encoding process enables the subsequent Diffusion Transformer to effectively ingest and interpret microsecond level motion cues that would otherwise remain entirely inaccessible to the system. This specific integration is of paramount importance because traditional shutter based RGB sensors are fundamentally constrained by their fixed integration intervals, a limitation that frequently results in the permanent loss of structural informational entropy through the manifestation of severe motion blur. Through the utilization of these dense tensors as a continuous and conditional guiding signal, the generative framework can precisely localize the trajectories of fast-moving boundaries and reconstruct intricate geometric textures with high fidelity. Consequently, this ensures that the synthesized intermediate frames are both physically grounded in real world radiance shifts and visually sharp, successfully overcoming the fundamental sampling bottlenecks that typically compromise standard video interpolation frameworks.

3.3 Event-Guided Diffusion Transformer

The architectural framework developed in this research is fundamentally rooted in the paradigm of Latent Diffusion Models, a sophisticated class of generative architectures that perform the intricate task of video frame interpolation within a highly compressed latent manifold. By shifting the generative process from the high dimensional and often redundant pixel domain to a more compact latent representation, the model can focus its representational capacity on the underlying semantic and structural essence of the scene. As illustrated in the comprehensive system diagram in Fig. 3(b), we utilize a frozen image encoder $\mathcal{E}_I$ [11], which is typically a pre trained component of a variational autoencoder, to project the discrete input anchor frames I_0 and I_1 into their respective latent counterparts, specifically $Z_0 = \mathcal{E}_I(I_0)\ and\ Z_1 = \mathcal{E}_I(I_1)$. This initial projection step is not merely a dimensionality reduction technique but a critical transformation that distills the visual information into a latent space where motion patterns and geometric consistencies are more readily discernable. Operating within this latent domain provides a dual advantage, as it drastically curtails the computational overhead required for the subsequent diffusion steps and allows the model to learn complex inter frame motion and structural dependencies with a level of abstraction that is difficult to achieve when dealing with raw pixel values. The operational efficiency gained through latent space processing is substantial, as it permits the use of deeper and wider transformer architectures within the same memory constraints, thereby significantly improving the overall scalability of the system.

At the heart of the generative process lies the denoising engine, which is composed of a series of enhanced Diffusion Transformer blocks as meticulously detailed in the architectural breakdown in Fig. 3(c). The transition from standard convolutional backbones to transformer-based architectures represents a significant advancement in the ability of the model to handle long range spatial relationships and global context. While the traditional Diffusion Transformer architecture utilizes self-attention as its primary mechanism for modeling spatial interactions within the latent patches, we have implemented a significant modification by introducing an additional Multi Head Cross Attention layer. This layer is purposefully designed to integrate the asynchronous motion evidence provided by event-based sensors into the spatial denoising trajectory. Within this Multi Head Cross Attention framework, the latent patches derived from the primary image features function as queries that interact with the event-based condition features v_0 and v_1. These event features, which represent the high frequency temporal changes occurring during the integration time of the frames, serve as the keys and values in the attention operation. This cross-modal interaction allows the transformer to dynamically weigh the importance of specific motion cues at different spatial locations, ensuring that the synthesized details are consistent with the physical reality of the scene. By querying the event data, the transformer blocks can accurately determine the position of moving edges and textures at any given timestamp, which effectively bridges the chasm between synchronous frames and asynchronous events.

This innovative integration mechanism ensures that the generated textures and structural boundaries are precisely aligned with the microsecond level temporal trajectories captured by the event stream, resolving ambiguities caused by severe motion blur. In high-speed scenarios where traditional RGB sensors suffer from significant information loss, these event guided anchors provide the necessary physical evidence to resolve the profound uncertainties. Consequently, the model is able to synthesize intermediate frames that exhibit sharp structural boundaries and fine textual details, effectively overcoming the ghosting artifacts that plague conventional interpolation methods. Following the rigorous optimization methodology established in the standard pipeline [32], the ensemble of enhanced Diffusion Transformer blocks Φ_θ is trained through a supervised objective to accurately approximate the time dependent velocity field v_θ. This velocity field is a fundamental component of the flow matching objective, as it describes the deterministic and continuous transformation required to map a simple Gaussian noise distribution onto the complex and structured target latent distribution. This process involves the solution of a probability flow ordinary differential equation, which ensures that the transport of probability mass is smooth and well defined. By learning to predict this velocity field, the model gains the ability to reconstruct the latent path of the intermediate frame with high precision, ensuring that the final output is both perceptually realistic and mathematically grounded in the provided multimodal evidence. This process is formulated as:

$$v_\theta(Z_t, t) = \Phi_\theta(Z_t, \mathrm{Concat}(Z_0, Z_1), v_0, v_1, t), \quad (2)$$

where Z_t represents the noisy latent at diffusion timestep t, and $\mathrm{Concat}(Z_0, Z_1)$ serves as the spatial condition. During inference, we synthesize the intermediate frame by solving the probability flow ordinary differential equation (PF ODE) starting from random noise. In the operational phase of model inference, the generation of the novel intermediate frame is conceptualized as a systematic traversal through the latent manifold, which is achieved by numerically solving the probability flow ordinary differential equation. This complex process commences with an initial sample of pure Gaussian noise that represents the maximum entropy state of the stochastic trajectory. By leveraging the learned score function or velocity field encoded within the Diffusion Transformer backbone, the framework progressively refines this chaotic input to reveal the underlying structural information. To navigate this differential landscape with both mathematical precision and computational tractability, we integrate a first order Euler sampling scheme as the primary numerical solver for the underlying ordinary differential equation. This iterative approach partitions the continuous time trajectory into a sequence of discrete steps, where at each interval, the model predicts the local velocity field to guide the latent vector toward the most likely regions of the data distribution. Such a methodology ensures a robust and stable reconstruction of the entire latent path, effectively balancing the necessity for high fidelity synthesis with the practical requirements of inference speed and stability across diverse motion scenarios. Upon the successful completion of the denoising procedure, the resulting optimized latent representation serves as a compressed descriptor of the synthesized motion state. The final and visually coherent intermediate frame I_τ is subsequently reconstructed by projecting this latent vector back into the pixel domain through the application of the frozen image decoder $\mathcal{D}$, ensuring that the abstract features are translated into the sharp boundaries and intricate textures that characterize real world video sequences.

3.4 Motion-Aware Training Strategy

Conventional training paradigms utilized in the domain of computational video interpolation generally operate on the foundational assumption of spatial homogeneity; consequently, the objective function evaluates the reconstruction error of every pixel with an identical weighting factor regardless of its local motion characteristics. This uniform treatment is frequently insufficient for capturing the intricate nuances of high-speed environments because the critical informational content and the most prominent reconstruction errors are disproportionately concentrated within dynamic regions rather than the redundant static background. In these challenging scenarios, the primary objective of the neural network is to resolve the high frequency details and structural boundaries of fast-moving objects, a task that is often hampered by the presence of severe motion blur and temporal aliasing that standard unweighted loss functions fail to address with sufficient precision. To circumvent these fundamental limitations, we propose a novel and robust motion aware training strategy that purposefully exploits the inherent temporal sparsity and microsecond level precision of asynchronous event signals to focus the network optimization on these vital spatial temporal manifolds. By leveraging the unique property that event cameras only record per pixel luminance fluctuations, our approach naturally filters out the irrelevant static portions of the scene and concentrates the entire learning effort of the model on the specific areas where motion evidence is most abundant and motion blur is most prevalent. The inaugural and essential step in this sophisticated training pipeline is the construction of a binary motion mask M, which is achieved by systematically identifying spatial locations that demonstrate a significant density of event triggers over the inter frame interval, thereby creating a precise and physically grounded spatial guide for the subsequent weighted optimization process. This mask allows the model to distinguish between dynamic and static regions:

$$M = \begin{cases} 1, & if\ \sum_{k=0}^{B-1} |V(k)| > \gamma \\ 0, & otherwise. \end{cases} \tag{3}$$

where V is the event voxel grid and γ is a predefined threshold. To match the compressed latent space, the mask $M \in \mathbb{R}^{H \times W}$ is then downsampled by $16\times$ to define the weight map $M^{'} \in \mathbb{R}^{\frac{H}{16} \times \frac{W}{16}}$. Based on this, we define a weight map $W = \omega_m M^{'} + \omega_s\left(1 - M^{'}\right)$. Here, ω_m and ω_s represent the training weights for moving and static regions, respectively. By setting $\omega_m > \omega_s$, we emphasize the reconstruction of dynamic content. This weight is applied to the velocity-based diffusion loss, forcing the model to prioritize the reconstruction of fast-moving structures:

$$\mathcal{L}_{\text{velocity}}(\theta) \coloneqq \mathbb{E}_{Z_{gt},\epsilon,t}[||W \odot (v_\theta(Z_t, t) - \dot{\alpha}_t Z_{GT} - \dot{\sigma}_t \epsilon)||^2]. \tag{4}$$

Here, v_θ represents our model's predicted velocity field, Z_{gt} is the ground-truth latent, and $\epsilon \sim \mathcal{N}(0, \mathrm{I})$ is the standard Gaussian noise. The terms $\dot{\alpha}_t$ and $\dot{\sigma}_t$ define the time-dependent trajectory of the diffusion process, while Z_t is the noisy latent at timestep t. With $\odot$ denoting the element-wise multiplication, this weighted objective ensures that the learning process is concentrated on high-motion areas while maintaining stability in static regions. To guarantee that the synthesized intermediate frames exhibit a rigorous degree of high level semantic coherence and conceptual integrity, we integrate a specialized objective known as the Representation Alignment loss, which is denoted as the REPA loss [32]. This strategic addition is motivated by the inherent tendency of generative diffusion processes to occasionally produce visually plausible but semantically inconsistent artifacts when guided solely by low level reconstruction errors. The fundamental operational principle of this alignment objective involves the synchronization of the internal hidden states of the Diffusion Transformer with the highly robust and abstract feature representations extracted from a fixed, pre trained self-supervised vision model. To facilitate this cross architectural alignment, we employ a trainable multi-layer perceptron projection head that functions as a bridge between the latent space of the generator and the embedding space of the teacher model. This projection mechanism is crucial for reconciling the disparate dimensionalities and stylistic variances that exist between the generative features and the discriminative semantic priors. By mathematically maximizing the cosine similarity between these two distinct sets of representations, the optimization process exerts a strong regularizing influence on the generative backbone. This objective forces the model to prioritize the orientation of its feature vectors in a manner that aligns with established visual concepts, thereby encouraging the production of noise invariant and semantically dense features that act as a reliable navigational guide for the iterative reconstruction process. This semantic grounding ensures that the final output is not only sharp at the pixel level but also logically consistent at the object level, effectively preventing the model from generating erroneous or physically impossible interpolations in challenging dynamic scenarios. The final training objective is a weighted combination of the motion and alignment components:

$$\mathcal{L} = \mathcal{L}_{\text{velocity}} + \lambda \mathcal{L}_{\text{REPA}}, \tag{5}$$

where λ is a hyperparameter balancing motion accuracy and perceptual consistency. This strategy ensures the framework remains physically grounded while maintaining sharp visual details across the entire frame.

4. Experiments

4.1 Experimental Setup

4.1.1 Implementation details

The architectural foundation of the proposed framework is centered on the SiT-XL/2 [32] variant of the Scalable Interpolation Transformer. This specific backbone is selected due to its demonstrated efficiency in handling high resolution video synthesis while maintaining a manageable computational footprint. To further optimize the processing requirements and enhance the scalability of the training process, the entire denoising and interpolation procedure is executed within a highly compressed latent manifold. Specifically, we utilize a spatial downsampling factor of 16, which significantly reduces the dimensionality of the feature maps. This compression allows the multi head attention layers of the Diffusion Transformer to operate on more compact representations without losing critical structural information, effectively balancing the computational load with the need for high fidelity reconstruction.

The software implementation of our model is realized using the PyTorch deep learning library, providing the necessary flexibility for custom layer development and efficient gradient computation. All experiments and training iterations are conducted on an NVIDIA A100 graphics processing unit, leveraging its large memory capacity and high throughput to facilitate stable training at the desired resolution. We employ the AdamW optimizer, which incorporates decoupled weight decay to enhance the generalization capabilities of the transformer weights and prevent overfitting during the later stages of training. The training duration is fixed at a total of 100 epochs, a timeframe that has been empirically verified to allow for the full convergence of both the motion aware velocity loss and the semantic representation alignment objectives.

To maintain a balance between stochastic gradient noise and training stability, a global batch size of 32 is utilized across the training iterations. The optimization starts with an initial learning rate of 10^{-4}, providing sufficient initial step sizes to explore the loss landscape effectively during the beginning of the training process. To ensure a smooth transition toward fine grained weight adjustment and to secure a stable convergence, the learning rate is modulated via a cosine annealing schedule. This schedule gradually reduces the learning rate until it reaches a terminal value of 10^{-6} at the conclusion of the 100th epoch, effectively preventing unwanted oscillations as the model approaches the local minima.

In the event processing pipeline, the raw asynchronous event stream is discretized into $B = 8$ temporal bins to form the structured input voxel grids. This specific discretization level is chosen to preserve high frequency motion evidence while keeping the memory overhead within acceptable limits for real time processing potential. For the generation of the motion masks used in our specialized training strategy, we apply a threshold $\gamma = 0.5$ to the accumulated intensities within the event voxel grid. This threshold ensures that the model effectively distinguishes between relevant dynamic pixels and low intensity noise inherent in event-based data. Furthermore, to explicitly prioritize the reconstruction of fast-moving objects, the training weight ω_m for identified dynamic regions is set to 2.0, while the weight ω_s for static backgrounds remains at 1.0. This two-fold weighting scheme forces the Diffusion Transformer backbone to allocate more representational capacity to the complex motion areas where traditional RGB sensors suffer most severely from information loss.

4.1.2 Training and evaluation

For the comprehensive optimization of the proposed network, we utilize the Vimeo90k Septuplet dataset [33]. This large-scale benchmark provides a vast collection of video sequences containing diverse motion patterns and complex textures, which is essential for training high-capacity generative models [34]. Since native event camera data is not readily available for these sequences, we generate synthetic event streams through the ESIM simulator [34]. This simulation process accurately models the asynchronous nature of event cameras by converting standard intensity changes into a continuous stream of events, providing the model with the necessary temporal evidence to handle high speed scenarios. This conversion is critical for ensuring the model learns to associate discrete visual appearances with the continuous temporal dynamics provided by the event sensors.

To bolster the generalization capabilities of our framework and prevent the model from overfitting to specific spatial arrangements, we implement an extensive data augmentation protocol. This strategy involves applying random spatial cropping to extract 256×256 pixel patches, which ensures that the network is exposed to various local motion configurations. Additionally, we perform random horizontal and vertical flipping along with random rotations to increase the geometric diversity of the training samples. Regarding the loss function configuration, we maintain the hyperparameter settings for the Representation Alignment loss in strict accordance with the default parameters established in the original SiT study [32]. This alignment objective ensures that the latent features generated by our model are semantically consistent with those from pre trained vision models, thereby enhancing the overall perceptual fidelity of the synthesized images [35].

The experimental evaluation is performed on the widely recognized Middlebury [36] and SNU FILM [37] benchmarks. All quantitative assessments and qualitative comparisons are conducted at the precise center timestamp ($t = 0.5$) relative to the two consecutive input frames. This specific temporal location is selected as the evaluation point because it typically exhibits the largest displacement and highest uncertainty, making it the most rigorous test for interpolation accuracy. To provide a multifaceted analysis of our results, we employ a suite of metrics focused on different aspects of image quality. We primarily utilize Learned Perceptual Image Patch Similarity and its temporal variant, Flow Learned Perceptual Image Patch Similarity, to evaluate the visual realism and the temporal smoothness of the interpolated frames. Furthermore, we report the Peak Signal to Noise Ratio and the Structural Similarity Index to provide a standardized measurement of the pixel level reconstruction fidelity.

During the final inference process, the intermediate frames are synthesized using an iterative denoising approach. We employ a first order Euler sampler that solves the underlying probability flow ordinary differential equation through 100 discrete sampling steps. This configuration was chosen to ensure a high degree of convergence in the generative process, resulting in frames that maintain sharp structural boundaries and fine textual details. By utilizing 100 steps, the model effectively balances computational efficiency with the need for high quality visual outputs, particularly in dynamic scenes where traditional regression-based methods fail to produce sharp results.

4.2 Performance Evaluation

To rigorously validate the efficacy of the proposed Event Guided Diffusion Transformer, a comprehensive comparative analysis is conducted against a selection of prominent state of the art interpolation frameworks on the Middlebury [36] and SNU FILM [37] benchmarks, as documented in Table 1. The evaluation focuses on perceptual fidelity and temporal smoothness, quantified through the Learned Perceptual Image Patch Similarity (LPIPS) and its flow grounded extension, FloLPIPS. These metrics are particularly relevant for generative architectures, as they have been shown to correlate more closely with the human visual system's sensitivity to structural integrity and motion continuity compared to traditional error-based calculations like PSNR.

On the Middlebury benchmark, the proposed methodology attains superior performance, yielding an LPIPS score of 0.013 and a FloLPIPS value of 0.028. Such empirical outcomes substantiate the premise that the integration of asynchronous event signals empowers the model to better preserve intricate structural details that are typically lost during rapid motion. The exceptionally low FloLPIPS score specifically highlights the attainment of smooth and physically plausible transitions between frames, suggesting that the model successfully bridges the temporal gaps inherent in standard shutter-based acquisitions by using microsecond level motion evidence as a guiding prior.

A similar trajectory of performance superiority is observed across the varied difficulty tiers of the SNU FILM dataset. While the proposed method remains highly competitive in the easy and medium settings, its comparative advantage becomes significantly more pronounced as the complexity of the motion scenarios increases. In the hard category, the model achieves an LPIPS of 0.053, which represents a notable enhancement over contemporary generative solutions such as MADIFF [13], which records a higher perceptual error of 0.058. This trend of robust performance persists into the extreme motion sequences, where our framework consistently outshines other latent based diffusion models by maintaining perceptual clarity even under conditions of massive pixel displacement.

These results collectively confirm that event guidance allows the Diffusion Transformer to resolve severe motion ambiguities that frequently lead to catastrophic failures and artifacts in traditional RGB only models. As demonstrated in the qualitative comparisons presented in Figure 3 and Figure 4, the proposed approach successfully reconstructs well defined boundaries and sharp fine textures even in high-speed scenes. In stark contrast, several baseline methods suffer from debilitating ghosting artifacts and structural smearing, failures that are effectively mitigated by the precise temporal anchors provided by our event guided synthesis pipeline. This balance between mathematical reliability and visual sharpness is particularly critical for applications in autonomous systems where accurate digital perception is a technical necessity.

Table 1. Quantitative results (LPIPS/FloLPIPS, the lower the better) on test datasets.

Methods	Middlebury	SNU-FILM			
		easy	medium	hard	extreme
	LPIPS↓/FloLPIPS↓	LPIPS↓/FloLPIPS↓	LPIPS↓/FloLPIPS↓	LPIPS↓/FloLPIPS↓	LPIPS↓/FloLPIPS↓
MCVD	0.123/0.138	0.199/0.230	0.213/0.243	0.250/0.292	0.320/0.385
VFIformer	0.015/0.024	0.018/0.029	0.033/0.053	0.061/0.100	0.119/0.185
LDMVFI	0.019/0.044	0.014/0.024	0.028/0.053	0.060/0.114	0.123/0.204
MADIFF	0.016/0.034	0.013/0.021	0.025/0.048	0.058/0.110	0.125/0.210
Ours	0.015/0.032	0.011/0.021	0.023/0.045	0.053/0.105	0.122/0.202

Table 2 reports the performance of our method in terms of PSNR and SSIM. Although generative frameworks are often optimized for perceptual realism, our approach maintains competitive pixel-level accuracy across all tested benchmarks.

Fig 3. Qualitative comparison with state-of-the-art methods on the SNU-FILM dataset.

Fig 4. Qualitative comparison with state-of-the-art methods on the Middlebury dataset.

Our model achieves the best results on the SNU-FILM easy and medium categories, reaching a PSNR of 38.920 and 34.101 respectively. The advantage of event guidance is most prominent in the extreme motion scenario. Our framework achieves a PSNR of 23.951, outperforming other diffusion-based models such as LDMVFI [11] and MADIFF [13]. This gain shows that the high-frequency information in event streams provides essential anchors for pixel reconstruction in challenging scenes where standard frames are heavily blurred. These results confirm that our method effectively balances structural accuracy with the enhanced visual fidelity observed in our perceptual evaluations.

Table 2. Quantitative results (PSNR/SSIM, the higher the better) on test datasets.

Methods	Middlebury	SNU-FILM			
		easy	medium	hard	extreme
	PSNR↑/SSIM↑	PSNR↑/SSIM↑	PSNR↑/SSIM↑	PSNR↑/SSIM↑	PSNR↑/SSIM↑
LDMVFI	34.230/0.974	38.890/0.988	33.975/0.971	29.144/0.911	23.349 0.827
MADIFF	34.002/0.973	38.674/ 0.987	33.996/0.970	28.547/0.917	23.934/0.837
Ours	34.123/0.971	38.920/0.989	34.101/0.970	29.192/0.905	23.951/0.837

Table 3. Ablation study of key components on the SNU-FILM (extreme) dataset.

Event Fusion Method	$\mathcal{L}_{\text{velocity}}$	Motion-aware $\mathcal{L}_{\text{velocity}}$	$\mathcal{L}_{\text{REPA}}$	SNU-FLIM (extreme) LPIPS↓/FloLPIPS↓/PSNR↑/SSIM↑
Concat	√			0.142/0.237/23.645/0.824
Concat		√		0.130/0.230/23.702/0.828
Concat		√	√	0.128/0.222/23.720/0.831
MHCA	√			0.129/0.215/23.650/0.828
MHCA		√		0.124/0.209/23.880/0.834
MHCA		√	√	0.122/0.202/23.951/0.837

4.3 Ablation Study

To systematically evaluate the individual contributions of each proposed module, we perform a comprehensive ablation study using the SNU FILM extreme dataset. The quantitative results for various configurations are meticulously detailed in Table 3. This rigorous analysis focuses on the relative efficacy of the event fusion mechanisms, the specialized motion aware training objective, and the semantic representation alignment loss. By isolating these components, we can ascertain how each innovation specifically contributes to the overall enhancement of reconstruction fidelity and perceptual realism in high-speed scenarios.

A primary focus of this study is the comparison between a naive feature concatenation method and our sophisticated Multi Head Cross Attention (MHCA) mechanism. When employing the standard velocity loss, the concatenation baseline results in an LPIPS of 0.142 and a FloLPIPS of 0.237. Upon substituting this approach with the MHCA module, these metrics immediately improve to 0.129 and 0.215 respectively. This transition represents a significant reduction in temporal consistency error. Such empirical outcomes indicate that the attention-based paradigm is intrinsically better suited for synchronizing asynchronous event signals with spatial latent patches during the denoising process. Unlike static concatenation, which merely stacks feature channels, the MHCA layers allow the network to dynamically query relevant motion evidence, thereby resolving temporal aliasing more effectively.

The integration of the motion aware training strategy further bolsters the accuracy of the model. When the motion aware velocity loss is applied to the MHCA framework, the PSNR rises from 23.650 to 23.880, and the LPIPS value drops from 0.129 to 0.124. A similar positive trend is visible in the concatenation-based variants, where the motion aware loss improves the LPIPS from 0.142 to 0.130. These significant gains confirm that prioritizing high motion areas helps the model resolve structural ambiguities that traditional loss functions typically overlook. By concentrating the representational capacity of the model on spatial locations with significant event activity, the framework can more successfully reconstruct fast moving structures that are otherwise obscured by motion blur.

Finally, the addition of the Representation Alignment (REPA) loss serves as the final refinement stage for achieving the highest perceptual quality. Our full model configuration attains the best overall performance, reaching an LPIPS of 0.122, a FloLPIPS of 0.202, and a PSNR of 23.951. Comparing the initial MHCA baseline to this final version reveals a total PSNR gain of approximately 0.3 decibels and an SSIM increase from 0.828 to 0.837. This holistic improvement validates that semantic alignment provides the necessary constraints for generating sharp textures and maintaining high level consistency. Each component is therefore demonstrated to be a vital element of the interpolation pipeline, and their collective synergy ensures high quality video synthesis in the most demanding dynamic environments.

5. Conclusions

5.1 Summary

In summary, this comprehensive research documentation has introduced a sophisticated Event Guided Diffusion Transformer framework specifically engineered to address the persistent challenges of video frame interpolation within highly dynamic and high-speed operational environments. The core of this methodology lies in the synergistic integration of asynchronous, high frequency temporal evidence harvested from event-based sensors with the robust generative potential of the Diffusion Transformer backbone. Traditional interpolation strategies, which rely exclusively on standard shutter based RGB only sensors, frequently struggle when faced with rapid object trajectories or significant camera ego motion. Such technological limitations often manifest as debilitating motion blur or severe ghosting artifacts, both of which fundamentally compromise the structural integrity and visual quality of the synthesized frames. Our proposed approach successfully circumvents these obstacles by utilizing the continuous stream of event signals as a precise temporal guide, ensuring that the generative process remains anchored to the true physical motion occurring between discrete exposure intervals.

The selection of the Diffusion Transformer as the generative engine represents a strategic move away from the spatial constraints and inductive biases of conventional convolutional neural networks. By leveraging global self-attention mechanisms within a compressed latent space, the framework possesses the capacity to model complex, long range spatial temporal dependencies that are often invisible to local receptive fields. When coupled with the microsecond level motion cues provided by event cameras, the model is equipped with a dense record of luminance changes that effectively fills the informational voids present in standard video frames. This integration allows the system to reconstruct non-linear motion paths with a level of precision that was previously unattainable for purely frame based systems. The result is a framework that does not merely perform a statistical guess of the intermediate state but rather reconstructs it based on high resolution physical evidence.

Extensive empirical evaluations conducted on the widely recognized Middlebury and SNU FILM benchmarks provide compelling evidence regarding the effectiveness and robustness of the proposed framework. The experimental results demonstrate that our methodology consistently achieves an optimal equilibrium between objective pixel level accuracy and subjective perceptual realism. In rigorous testing scenarios characterized by extreme motion and complex textures, our framework has demonstrated a superior ability to preserve sharp structural boundaries and fine textural details. By outperforming several state of the art diffusion based baselines, the study confirms that the addition of event data acts as a critical regularizer for the generative process. This prevents the over smoothing typical of traditional regression-based methods while simultaneously avoiding the hallucinatory artifacts that can sometimes plague unconditioned generative models.

Two primary technical innovations underpin these significant performance gains: the implementation of the Multi Head Cross Attention mechanism and the development of a specialized motion aware training strategy. The MHCA module facilitates a more effective fusion of multimodal data by allowing the spatial latent tokens to query relevant motion features from the event voxel grid in a dynamic and spatially adaptive manner. Parallel to this, the motion aware training strategy utilizes a weighted loss function to concentrate the model's learning capacity on specific areas of high dynamic activity. By prioritizing the reconstruction of fast-moving objects over static backgrounds, this strategy ensures that the network is optimized to resolve the most challenging regions of the video sequence. Collectively, these contributions represent a significant advancement in the field of computer vision, establishing a new standard for high fidelity video interpolation in challenging and unpredictable scenarios.

5.2 Future Research Directions

While the empirical evidence presented in this thesis confirms the efficacy of the proposed Event Guided Diffusion Transformer, the study also highlights several critical areas that warrant further scholarly investigation. The transition from a research prototype to a production ready system necessitates addressing the fundamental bottlenecks associated with generative modeling and asynchronous sensor fusion. This evolution involves not only the refinement of existing algorithms but also the exploration of entirely new paradigms for cross modal information exchange between disparate visual sensors.

A primary objective for subsequent research involves the substantial optimization of inference efficiency within the diffusion framework. At present, the iterative denoising process inherent to Diffusion Transformers requires multiple sequential sampling steps, which translates to a significant computational burden that precludes real time execution on resource constrained edge devices. Future efforts will focus on the implementation of advanced acceleration techniques such as progressive distillation or the adoption of consistency models which can reduce the sampling requirement to a single step without compromising visual fidelity. Beyond algorithmic acceleration, exploring hardware aware model quantization and pruning strategies tailored for transformer architectures will be essential for deploying these models in time critical applications such as mobile robotics and handheld consumer electronics. This includes investigating how reduced precision affects the sensitive denoising trajectory and the structural integrity of the synthesized latent features [38].

Another promising avenue for exploration lies in the development of more efficient and descriptive event representations that transcend the limitations of structured voxel grids. While voxelization provides a convenient bridge to traditional neural networks, it inherently introduces quantization errors and fails to fully exploit the microsecond level temporal precision and extreme sparsity of the raw event stream. Future studies should investigate the use of point based or graph-based architectures that can process events as discrete geometric entities in a continuous coordinate space. By utilizing Spiking Neural Networks or asynchronous graph convolution frameworks, it may be possible to preserve the high-resolution temporal dynamics of the sensor more effectively while simultaneously reducing the memory footprint and the computational latency of the fusion module. Such architectures could potentially facilitate an event by event processing pipeline, significantly reducing redundant operations in static scene regions [39].

Furthermore, the robustness of the model must be extended to encompass a more diverse range of challenging environmental conditions. While the current framework excels in standard dynamic scenarios, its performance under extreme low light or high dynamic range conditions remains a subject for further validation. Given that event cameras possess a significantly higher dynamic range than traditional CMOS sensors, there is a clear opportunity to develop specialized conditioning mechanisms that leverage this physical advantage to reconstruct frames in scenes with severe overexposure or deep shadows. Future research will specifically examine the integration of exposure aware priors and neuromorphic noise filters to enhance the signal to noise ratio in adversarial lighting environments. Such advancements would broaden the applicability of the system in autonomous navigation and aerospace engineering, where reliable visual perception is required regardless of the ambient lighting conditions [40].

Finally, we plan to investigate the extension of this framework from single frame interpolation to multi frame video generation and long-term prediction. Current VFI methods primarily focus on the midpoint between two frames, but the dense temporal information from events could theoretically support the synthesis of arbitrary frame rates or even the extrapolation of future motion trajectories. This expansion would involve the design of recurrent or autoregressive transformer blocks capable of maintaining temporal consistency over longer sequences. Ultimately, the evolution of this research will focus on creating a more unified and efficient perception system that seamlessly blends the generative power of modern transformers with the unique physical properties of neuromorphic sensors. The pursuit of these research directions will be instrumental in realizing the next generation of digital perception systems that are both computationally efficient and physically grounded in the complexities of the real world.

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